The Effective Active-Face Count Conflates Redundancy with Dimension

An empirical rank-collapse finding for the quartic DLE Bellman fibers

August 28, 2026

Abstract

The commentary on latent Bellman fibers introduced a soft effective active-face count $N_{\mathrm{eff}}(a)$ as a diagnostic for the local geometric complexity of the polyhedral defect function d_r. We report a direct numerical test of this diagnostic on cut pools produced by a working head–tail lazy-separation solver for the concrete quartic instance. At the certified root $a_r^\star$ (where $d_r(a_r^\star) = 0$), N_{eff} is large and grows sharply with r, while at generic interior points of the simplex it remains small. We show this growth is a redundancy artifact, not a dimension artifact: the numerical rank of the ε-near-active cut set saturates at M (the ambient affine dimension) using a small fraction of the cuts that N_{eff} weights, and the saturation rank does not grow with r. We propose a rank-based refinement of the diagnostic and state the corresponding open question precisely. The finding is based on two converged data points ($r = 2, 3$) and should be read as a diagnostic clarification, not a resolution of Problem 6.

1 Setup and notation

We recall the relevant objects from the latent-fibers commentary. Fix M, the simplex Δ_M, a target law $B_\star$, and the recursive nonnegative defect

$$d_{r+1}(a) = \min_{b^{(0)},\dots,b^{(4)} \in \Delta_M} \left[\sum_{x=0}^{4} P(x)\, d_r\big(b^{(x)}\big) + \lambda \Big\| Q * a - \sum_{x=0}^{4} P(x)\, \delta_x * b^{(x)} \Big\|_1 \right], \qquad \lambda > 0,$$

with $d_r(a) = 0 \iff a \in \mathcal{C}_r(B_\star)$. For fixed finite M, each d_r is convex and polyhedral,

$$d_r(a) = \max_{\alpha \in A_r} \big\{ g_{r,\alpha}^\top a - h_{r,\alpha} \big\}.$$

Given a candidate a, write $c_\alpha(a) = g_{r,\alpha}^\top a - h_{r,\alpha}$, $\hat{d}_r(a) = \max_\alpha c_\alpha(a)$, and the facet gap $\Delta_\alpha(a) = \hat{d}_r(a) - c_\alpha(a) \ge 0$. For a scale $\tau > 0$,

$$w_\alpha(a) = \exp\!\left(-\frac{\Delta_\alpha(a)}{\tau} \right), \qquad p_\alpha(a) = \frac{w_\alpha(a)}{\sum_\beta w_\beta(a)}, \qquad N_{\mathrm{eff}}(a) = \frac{1}{\sum_\alpha p_\alpha(a)^2}.$$

Recall $1 \le N_{\mathrm{eff}}(a) \le |A_r|$, with $N_{\mathrm{eff}}(a) \ll |A_r|$ interpreted as head-dominated (compressible) local geometry and $N_{\mathrm{eff}}(a) \asymp |A_r|$ as diffuse geometry.

2 Concrete instance and solver

We use the quartic pair $P = (\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8})$, $Q = (\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64})$, support cutoff $M = 8$, and the certified three-atom target

$$B_8 = 0.53562333995590705\, \delta_0 + 0.42450840479209673\, \delta_1 + 0.039868255251995686\, \delta_2,$$

so that $c_0 = 0.53562333995590705$, $c_1 = 0.96013174474800378$. Cut pools A_r were produced by an adaptive head–tail lazy-separation solver implementing the recursion above together with its Fenchel-dual Bellman step, run with $\lambda = 1$. The solver reproduced the independently-verified regression values

r	$D_r(B_8)$ (solver)	outer rounds	$\lvert A_r \rvert$
1	0.562605012645	8	9
2	0.589547223565	20	1295
3	0.601704229323	34	2611
4	0.607608579698	23	211*

matching the previously certified monolithic values to within 10^{-6} or better at every level. *Level 4 was not fully matured at the time of this snapshot: the solver was mid-computation on level 5 when the state was checkpointed, so A_4 should be read as a partial pool, not a converged one; it is included only for context and is excluded from the rank analysis below. Let $a_r^\star$ denote the certified root achieving $D_r(B_8)$, i.e. $d_r(a_r^\star) = 0$ to the certification tolerance 10^{-7}.

3 The raw diagnostic: large and growing at the root

Table 1 reports N_{eff} evaluated at $a_r^\star$ for $\tau = 0.01$, alongside N_{eff} at five points drawn from a Dirichlet$(8, 6, 3, 1, 1, 1, 1, 1, 1)$ sample of the simplex (skewed toward small support, as is typical of the laws arising in this problem).

r	$\lvert A_r \rvert$	$N_{\text{eff}}(a_r^\star)$	N_{eff} at five random simplex points
1	9	6.4	1.0, 1.0, 1.0, 1.0, 1.0
2	1295	1115.9	22.9, 7.4, 7.8, 4.5, 4.7
3	2611	2073.5	5.3, 10.7, 8.9, 57.7, 53.3

Table 1: N_{eff} at the certified root versus at generic points, $\tau = 0.01$.

Read at face value, this is close to the pessimistic case flagged in the commentary: $N_{\text{eff}}(a_r^\star)$ tracks a large fraction of $\lvert A_r \rvert$ and grows sharply from $r = 1$ to $r = 3$, while away from the root the geometry is visibly head-dominated. Since $a_r^\star$ is exactly the point the outer master must certify at every level, this is also the point whose local complexity is operationally relevant to solver cost.

4 The rank test

We next partition the pool by facet gap at the root, $S_\varepsilon = \{\alpha : \Delta_\alpha(a_r^\star) < \varepsilon\}$, and compute the numerical rank of the matrix formed by stacking $\{g_{r,\alpha}\}_{\alpha \in S_\varepsilon}$ (tolerance 10^{-8}).

Observation 1 (Rank saturation). *At both tested levels, the rank of S_ε reaches its terminal value using between 2% and 6% of the cuts eventually admitted at $\varepsilon = 10^{-2}$, and does not increase further as $\lvert S_\varepsilon \rvert$ grows by more than an order of magnitude beyond that point.*

Remark 1 (Why 8, not 9). *$M = 8$, so $g_{r,\alpha} \in \mathbb{R}^9$ nominally, but every cut is evaluated only as an affine functional on the affine hull $\{a : \sum_j a_j = 1\}$, an 8-dimensional set. Two cuts whose g-vectors differ by a multiple of the all-ones vector induce the same functional on that hull. The terminal rank of 8 is therefore the ambient ceiling for this instance, not an emergent or coincidental number.*

r	ε	$\lvert S_\varepsilon \rvert$	$\operatorname{rank} G(S_\varepsilon)$
2	10^{-9}	1	1
2	10^{-4}	51	8
2	10^{-3}	206	8
2	10^{-2}	1063	8
3	10^{-9}	7	5
3	10^{-4}	98	8
3	10^{-3}	351	8
3	10^{-2}	1779	8

Table 2: Rank of the ε-near-active cut set as a function of threshold.

5 Interpretation

The saturation in Table 2 shows that N_{eff}, as defined by the softmax entropy formula, is dominated by clusters of cuts that are affinely dependent on a much smaller subset already present at looser thresholds. Once 8 independent directions are found, every further near-active cut is, up to numerical tolerance, a linear combination of directions already available. The diagnostic in Section 1.3 of the commentary conflates two distinct notions:

- the number of cuts within ε of the active envelope (what N_{eff} measures),
- the dimension of the space they span (what determines whether they carry independent information).

A pool can be simultaneously large, diffuse by the first measure, and low-dimensional by the second. That is what is observed here.

5.1 What this does and does not show

It does *not* show that Problem 6, in its original uniform-over-the-simplex form, has a positive answer: only two converged levels were tested, and S_ε reaching rank 8 is close to a necessary fact of the ambient dimension rather than a deep structural bound. It also does not resolve whether the trajectory-relevant complexity is bounded: the *cardinality* needed to reach full rank at the tightest informative threshold grew from 51 ($r = 2$) to 98 ($r = 3$), which is itself an unresolved trend on two points.

It *does* show that the entropy-weighted N_{eff} is the wrong instrument for the question the commentary poses, because it cannot distinguish "many independent facets" from "many redundant near-copies of few facets," and that in this concrete instance the second case is what actually occurs at the levels tested.

5.2 A sharper diagnostic and a sharper question

Define the *basis count* at threshold ε,

$$\beta_r(a, \varepsilon) = \min\big\{ \, |B| : B \subseteq S_\varepsilon(a), \ \operatorname{rank} G(B) = \operatorname{rank} G(S_\varepsilon(a)) \, \big\},$$

i.e. the size of a minimal spanning subset of the near-active cuts, computable in practice by a greedy or QR-pivoted extraction. Unlike N_{eff}, β_r is insensitive to duplication by construction.

Problem 9 (Basis-count growth)**.** *For the certified roots $a_r^\star$ along the Bellman trajectory, does $\beta_r(a_r^\star, \varepsilon)$, for fixed small ε, stay bounded in r, grow sub-linearly (e.g. logarithmically), or grow at a rate comparable to $|A_r|$ itself? The two available data points (*$51 \to 98$*) are consistent with more than one of these and do not settle the question.*

5.3 A practical corollary, if the redundancy is structural

If β_r turns out to be small relative to $|A_r|$ at every level along the trajectory, the practical corollary is immediate: a redundancy-aware pruning step — discarding any cut that lies, within numerical tolerance, in the affine span of already-retained near-active cuts — could shrink the pools driving the head–tail solver without loss of certification power. Given the observed slowdown when extending the same solver from $r = 4$ towards $r = 5$, and given that pool sizes of 1295 and 2611 at $r = 2, 3$ apparently carry no more than 8 independent directions near the point that actually matters, this is a concrete, testable candidate explanation for a nontrivial fraction of that slowdown, independent of any claim about the asymptotics of B_n itself.

6 Scope

This note reports a diagnostic finding on one solver run, one concrete (P, Q, B_8, M, λ) instance, and two converged levels. It refines rather than answers Problem 6 of the latent- fibers commentary, and says nothing by itself about m_n, $D_r(B_8)$ for $r \geq 5$, or the original quartic predecessor growth question. Its claim is narrow and, we believe, solid: *the specific entropy-weighted effective-face-count diagnostic, applied at the points that matter operationally, measures redundancy at least as much as it measures genuine facet complexity, in this instance.*